Umer Gupta

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PROFILE

My current work is focused on geometric deep learning - generative geometric modelling, symmetry-respecting architectures, and topological priors. With the University of Leipzig and ETH Zurich, I developed an autoregressive and diffusion-based generative methodology for branching morphologies like neurons of variable size under topological guidance. I am drawn to a broader cluster of biomedical problems where the underlying structure of the domain is central - drug design, single-cell biology, and connectomics.

PUBLICATIONS

Autoregressive Frontier Expansion: Growing Trees with Graph Machine Learning. **GRaM Workshop, ICLR 2026.**
Umer Gupta, Saku Peltonen, and Martin Ritzert
Poster presentation, Rio de Janeiro, April 2026.

Contributions: (i) An autoregressive process that jointly generates the topology (if nodes should expand) and 3D geometry (positions of new nodes) of branching structures; (ii) an SO(2)-equivariant EGNN using parent-centric local frames, suited to axis-aligned biological structures; (iii) diffusion-based generation at each step instead of one shot prediction; (iv) topological conditioning via 0-dimensional persistent homology under multiple filtrations. Demonstrated on botanical tree crowns; extensions to neuronal morphologies in follow-up work.

RESEARCH EXPERIENCE

Research Collaborator, ETH Zurich & University of Leipzig **Sep 2025 - Present**
Lead ML Engineer, New Gradient **Oct 2023 - Present**
Applied ML research and production work on geospatial problems in monitoring ecosystems health under UK research grants. Spin-out: Calterra.

- **Peatlands (IUK Grant).** Self-supervised geospatial backbone (BEiT/MAE) trained on over 20,000 sq. km. of multimodal UK peatland imagery (RGB + DSM) and downstream segmentation model (UPerNet) for delineating erosion features on ground. Engineered active-learning sample selection for domain generalisation across heterogeneous data sources. 10x mIoU improvement over the incumbent DEFRA-backed baseline.
- **Woodlands (UKSA Grant).** DINOv3 backbone with centerpoint, Mask2Former or RF-DETR, and Neural Field heads for tree counting, canopy segmentation, and species classification. Standardised a multi-dataset corpus of 76k tiles and over 1M tree annotations, unifying heterogeneous labels and resolutions.

3D subsurface modelling for precision exploration. Voxelised highly sparse, multimodal borehole logs (geochemistry and geophysics) and trained a Single-Stride Sparse Transformer optimised for non-empty voxels to perform 3D spatial inpainting and regression on masked mineral zones. Utilised the masked voxel regression to predict underground copper and gold grades. Awarded $4,000 for precision mining exploration innovation.

Origin Peptides (proteins). Predicted self-assembly indicators of peptides from sequence features and latent structural representations. Contributed to selection for Innovate UK's £6.4M SMMIP programme now active and working on using ML to inform experimental conditions for something protein synthesis.

TECHNICAL PROJECTS

Industry dissertation with Amazon (2023). Extended the SASRec sequential recommender from next-item to 14-day future temporal window prediction; window-aware variants improved Recall@10 by ~17.4% and NDCG@10 by ~25% on longer windows relative to autoregressive predictions with vanilla SASRec.
Code-mixed sentiment analysis (2022). Fine-tuned XLM-T and other multilingual transformers on Hinglish tweets. Explored latent mix-up - interpolating hidden representations within- vs across-class.
Cormeum Technologies (2021). Adaptive ECG/PPG signal-processing algorithms for in-flight health monitoring in noisy fighter-jet environments. Indian Ministry of Defence contract, collaboration with a Cornell team.

EDUCATION

2022 - 2023	**University of Edinburgh** MSc in Data Science | Distinction Relevant Courses: Machine Learning and Pattern Recognition, Machine Learning Theory, Reinforcement Learning.	**Edinburgh, UK**
2017 - 2020	**Sri Venkateswara College, Delhi University** Bachelor of Science (Hons) in Mathematics | GPA: 8.62 out of 10	**New Delhi, India**
2016 - 2017	**College of Arts and Sciences, Cornell University** 1st Semester, Undecided | GPA: 3.67 out of 4.3	**Ithaca, USA**

TECHNICAL FOCI

PyTorch, PyG; equivariant GNNs, geometric DL, diffusion generative models, masked-image and masked-3D pretraining, topological data analysis (persistent homology), large-scale geospatial ML, AI for biology.